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Lab 1: Implementation of Source code management using git commands through github

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OBJECTIVE:

Implementation of Source Code Management Using Git Commands Through GitHub.

AIM:

To understand and implement Source Code Management (SCM) using Git and GitHub for tracking changes, managing versions, and collaborating on software projects.

THEORY:

Source Code Management (SCM) is the practice of tracking and managing changes made to source code during software development. It helps maintain version history, supports collaboration, and allows previous versions of a project to be restored when needed.

Git is a distributed version control system used to record changes and manage different versions of a project. GitHub is an online platform that hosts Git repositories and enables developers to store, share, and collaborate on code. In this lab, Git commands are used to manage project versions and synchronize the local repository with GitHub.

OBJECTIVES

- Learn the basics of Git.
- Create and manage Git repositories.
- Track project changes using commits.
- Connect local repositories with GitHub.
- Push and pull project updates.
- Understand collaborative software development.

ALGORITHM

1. Install Git on the system.
2. Create a GitHub account.
3. Create a repository on GitHub.
4. Initialize a local Git repository.
5. Add project files.
6. Stage files using Git.
7. Commit changes.
8. Link local repository to GitHub.
9. Push files to GitHub.
10. Verify successful synchronization.

CODE

- Initialize Repository

```
git init
```

- Configure User Information

```
git config --global user.name "your name"
git config --global user.email "your@email.com"
```

- Check Repository Status

```
git status
```

- Add Files

```
git add .
```

- Commit Changes

```
git commit -m "Initial Commit"
```

- Connect Repository to GitHub

```
git remote add origin https://github.com/username/repository.git
```

- Push to GitHub

```
git branch -M main
git push -u origin main
```

- Pull Latest Changes

```
git pull origin main
```

OUTPUT

The Git repository was successfully initialized and connected to GitHub.

- Repository created successfully.
- Files tracked using Git.
- Changes committed successfully.
- Remote repository linked.
- Project pushed to GitHub.

RESULT

The implementation of Source Code Management using Git and GitHub was completed successfully. Version control operations such as repository creation, staging, committing, and synchronization with GitHub were performed successfully.

CONCLUSION

This experiment provided practical knowledge of Source Code Management using Git and GitHub. It demonstrated how version control systems help manage software projects efficiently by maintaining code history, enabling collaboration, and supporting Agile Software Development practices.